



Project 1

Prepared By:

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1) RTL Code:

Mux 4x1:

```
module mux4x1(I0, I1, I2, I3, op, out_mux4x1);
input [47:0] I0, I1, I2, I3;
input [1:0] op;
output [47:0] out_mux4x1;

assign out_mux4x1 = (op == 2'b00)? I0 : (op == 2'b01)? I1 : (op == 2'b10)? I2 :
I3;

endmodule
```

DFF + MUX 2x1:

```
module dff_mux(D, CLK, RST, Q, CE, out_dffmux, sel);
parameter SIZE = 18;
parameter RSTTYPE = "SYNC";
input [SIZE-1:0] D;
input CLK, RST, CE, sel;
output reg [SIZE-1:0] Q;
output [SIZE-1:0] out_dffmux;

generate
    if (RSTTYPE == "ASYNC") begin
        always @(posedge CLK or posedge RST) begin
            if (RST)
                Q <= 0;
            else if (CE)
                Q <= D;
        end
    end
    else begin
        always @(posedge CLK) begin
            if (RST)
                Q <= 0;
            else if (CE)
                Q <= D;
        end
    end
endgenerate

assign out_dffmux = (sel == 1)? Q : D;

endmodule
```

**Top:**

```
module dsp_top(A, B, D, BCIN, C, PCIN, OPMODE, CLK, CARRYIN, RSTA, RSTB, RSTM,
RSTP, RSTC, RSTD, RSTCARRYIN, RSTOPMODE, CEA, CEB, CEM ,CEP, CEC, CED, CECARRYIN,
CEOPMODE, BCOUT, PCOUT, P, M, CARRYOUT, CARRYOUTF);
parameter A0REG = 0;
parameter A1REG = 1;
parameter B0REG = 0;
parameter B1REG = 1;
parameter CREG = 1;
parameter DREG = 1;
parameter MREG = 1;
parameter PREG = 1;
parameter CARRYINREG = 1;
parameter CARRYOUTREG = 1;
parameter OPMODEREG = 1;
parameter CARRYINSEL = "OPMODE5";
parameter B_INPUT = "DIRECT";
parameter RSTTYPE = "SYNC";

input [17:0] A, B, D, BCIN;
input [47:0] C, PCIN;
input [7:0] OPMODE;
input CLK, CARRYIN, RSTA, RSTB, RSTM, RSTP, RSTC, RSTD, RSTCARRYIN, RSTOPMODE,
CEA, CEB, CEM ,CEP, CEC, CED, CECARRYIN, CEOPMODE;
output [17:0] BCOUT;
output [47:0] PCOUT, P;
output [35:0] M;
output CARRYOUT, CARRYOUTF;
wire [1:0] REG_01, REG_23, MUX_01, MUX_23;
wire REG_4, REG_5, REG_6, REG_7, MUX_4, MUX_5, MUX_6, MUX_7, CC_MUX, CYI_OUT,
CIN, PRE_CARRYO, CYO_OUT;
wire [17:0] D_REG, B0_REG, A0_REG, D_MUX, B0_MUX, A0_MUX, PAS_OUT, PAS_MUX,
B1_REG, B1_MUX, A1_REG, A1_MUX, BMUX;
wire [47:0] C_REG, C_MUX, ZMUX_OUT, XMUX_OUT, SUM, P_REG, P_MUX;
wire [35:0] MUL_OUT, M_REG, M_OUT;
wire [48:0] PAS2;

//OPMODES registers
dff_mux #(.RSTTYPE(RSTTYPE), .SIZE(2)) op01 (.D(OPMODE[1:0]), .CLK(CLK),
.RST(RSTOPMODE), .Q(REG_01), .CE(CEOPMODE), .out_dffmux(MUX_01),
.sel(OPMODEREG));
dff_mux #(.RSTTYPE(RSTTYPE), .SIZE(2)) op23 (.D(OPMODE[3:2]), .CLK(CLK),
.RST(RSTOPMODE), .Q(REG_23), .CE(CEOPMODE), .out_dffmux(MUX_23),
.sel(OPMODEREG));
```



```
dff_mux #(.RSTTYPE(RSTTYPE), .SIZE(1)) op4 (.D(OPMODE[4]), .CLK(CLK),
.RST(RSTOPMODE), .Q(REG_4), .CE(CEOPMODE), .out_dffmux(MUX_4), .sel(OPMODEREG));
dff_mux #(.RSTTYPE(RSTTYPE), .SIZE(1)) op5 (.D(OPMODE[5]), .CLK(CLK),
.RST(RSTOPMODE), .Q(REG_5), .CE(CEOPMODE), .out_dffmux(MUX_5), .sel(OPMODEREG));
dff_mux #(.RSTTYPE(RSTTYPE), .SIZE(1)) op6 (.D(OPMODE[6]), .CLK(CLK),
.RST(RSTOPMODE), .Q(REG_6), .CE(CEOPMODE), .out_dffmux(MUX_6), .sel(OPMODEREG));
dff_mux #(.RSTTYPE(RSTTYPE), .SIZE(1)) op7 (.D(OPMODE[7]), .CLK(CLK),
.RST(RSTOPMODE), .Q(REG_7), .CE(CEOPMODE), .out_dffmux(MUX_7), .sel(OPMODEREG));

//Stage 1 registers
dff_mux #(.RSTTYPE(RSTTYPE)) DM_D (.D(D), .CLK(CLK), .RST(RSTD), .Q(D_REG),
.CE(CED), .out_dffmux(D_MUX), .sel(DREG));
assign BMUX = (B_INPUT == "DIRECT")? B : (B_INPUT == "CASCADE")? BCIN : 0;
dff_mux #(.RSTTYPE(RSTTYPE)) DM_B0 (.D(BMUX), .CLK(CLK), .RST(RSTB), .Q(B0_REG),
.CE(CEB), .out_dffmux(B0_MUX), .sel(B0REG));
dff_mux #(.RSTTYPE(RSTTYPE)) DM_A0 (.D(A), .CLK(CLK), .RST(RSTA), .Q(A0_REG),
.CE(CEA), .out_dffmux(A0_MUX), .sel(A0REG));
dff_mux #(.RSTTYPE(RSTTYPE), .SIZE(48)) DM_C (.D(C), .CLK(CLK), .RST(RSTC),
.Q(C_REG), .CE(CEC), .out_dffmux(C_MUX), .sel(CREG));

//PAS1 + MUX
assign PAS_OUT = (MUX_6 == 1)? (D_MUX - B0_MUX) : (D_MUX + B0_MUX);
assign PAS_MUX = (MUX_4 == 1)? PAS_OUT : B0_MUX;

//B1
dff_mux #(.RSTTYPE(RSTTYPE)) DM_B1 (.D(PAS_MUX), .CLK(CLK), .RST(RSTB),
.Q(B1_REG), .CE(CEB), .out_dffmux(B1_MUX), .sel(B1REG));

//A1
dff_mux #(.RSTTYPE(RSTTYPE)) DM_A1 (.D(A0_MUX), .CLK(CLK), .RST(RSTA),
.Q(A1_REG), .CE(CEA), .out_dffmux(A1_MUX), .sel(A1REG));

assign BCOUT = B1_MUX;

//Multiplier stage
assign MUL_OUT = B1_MUX * A1_MUX;
dff_mux #(.RSTTYPE(RSTTYPE), .SIZE(36)) DM_M (.D(MUL_OUT), .CLK(CLK), .RST(RSTM),
.Q(M_REG), .CE(CEM), .out_dffmux(M_OUT), .sel(MREG));

//Buffer
genvar i;
generate
    for (i = 0; i < 36; i=i+1) begin : MBUF
        buf MBUFF (M[i], M_OUT[i]);
    end
end
```



```
endgenerate

//X MUX
mux4x1 X (.I0(48'b0), .I1({12'b0, M_OUT}), .I2(P_MUX), .I3({D_MUX[11:0], A1_MUX,
B1_MUX}), .op(MUX_01), .out_mux4x1(XMUX_OUT));

//Z MUX
mux4x1 Z (.I0(48'b0), .I1(PCIN), .I2(P_MUX), .I3(C_MUX), .op(MUX_23),
.out_mux4x1(ZMUX_OUT));

//CIN
assign CC_MUX = (CARRYINSEL == "OPMODE5")? MUX_5 : (CARRYINSEL == "CARRYIN")?
CARRYIN : 0;
dff_mux #(.RSTTYPE(RSTTYPE), .SIZE(1)) CYI (.D(CC_MUX), .CLK(CLK),
.RST(RSTCARRYIN), .Q(CYI_OUT), .CE(CECARRYIN), .out_dffmux(CIN),
.sel(CARRYINREG));

//PAS 2 STAGE
assign PAS2 = (MUX_7 == 1) ? ({1'b0, ZMUX_OUT} - ({1'b0, XMUX_OUT} + CIN)) :
({1'b0, ZMUX_OUT} + {1'b0, XMUX_OUT} + CIN);
assign SUM = PAS2[47:0];
assign PRE_CARRYO = PAS2[48];

//CARRYO
dff_mux #(.RSTTYPE(RSTTYPE), .SIZE(1)) CYO (.D(PRE_CARRYO), .CLK(CLK),
.RST(RSTCARRYIN), .Q(CYO_OUT), .CE(CECARRYIN), .out_dffmux(CARRYOUT),
.sel(CARRYOUTREG));
assign CARRYOUTF = CARRYOUT;

//P
dff_mux #(.RSTTYPE(RSTTYPE), .SIZE(48)) PFFMUX (.D(SUM), .CLK(CLK), .RST(RSTP),
.Q(P_REG), .CE(CEP), .out_dffmux(P_MUX), .sel(PREG));
assign P = P_MUX;
assign PCOUT = P_MUX;

endmodule
```



2) Testbench code:

```
module dsp_tb();
reg [17:0] A_tb, B_tb, D_tb, BCIN_tb;
reg [47:0] C_tb, PCIN_tb;
reg [7:0] OPMODE_tb;
reg CLK_tb, CARRYIN_tb, RSTA_tb, RSTB_tb, RSTM_tb, RSTP_tb, RSTC_tb, RSTD_tb,
RSTCARRYIN_tb, RSTOPMODE_tb, CEA_tb, CEB_tb, CEM_tb, CEP_tb, CEC_tb, CED_tb,
CECARRYIN_tb, CEOPMODE_tb;
wire [17:0] BCOUT_DUT;
wire [47:0] PCOUT_DUT, P_DUT;
wire [35:0] M_DUT;
wire CARRYOUT_DUT, CARRYOUTF_DUT;

dsp_top DUT (A_tb, B_tb, D_tb, BCIN_tb, C_tb, PCIN_tb, OPMODE_tb, CLK_tb,
CARRYIN_tb, RSTA_tb, RSTB_tb, RSTM_tb, RSTP_tb, RSTC_tb, RSTD_tb, RSTCARRYIN_tb,
RSTOPMODE_tb, CEA_tb, CEB_tb, CEM_tb, CEP_tb, CEC_tb, CED_tb, CECARRYIN_tb,
CEOPMODE_tb, BCOUT_DUT, PCOUT_DUT, P_DUT, M_DUT, CARRYOUT_DUT, CARRYOUTF_DUT);

initial begin
    CLK_tb = 0;
    forever
        #1 CLK_tb = ~CLK_tb;
end

integer i;
initial begin
//reset
    RSTA_tb = 1;
    RSTB_tb = 1;
    RSTM_tb = 1;
    RSTP_tb = 1;
    RSTC_tb = 1;
    RSTD_tb = 1;
    RSTCARRYIN_tb = 1;
    RSTOPMODE_tb = 1;
    A_tb = $random;
    B_tb = $random;
    D_tb = $random;
    BCIN_tb = $random;
    C_tb = $random;
    PCIN_tb = $random;
    OPMODE_tb = $random;
    CARRYIN_tb = $random;
    CEA_tb = $random;
    CEB_tb = $random;
```



```
CEM_tb = $random;
CEP_tb = $random;
CEC_tb = $random;
CED_tb = $random;
CECARRYIN_tb = $random;
CEOPMODE_tb = $random;
@(negedge CLK_tb);
if ((BCOUT_DUT == 0) && (PCOUT_DUT == 0) && (P_DUT == 0) && (M_DUT == 0) &&
(CARRYOUT_DUT == 0) && (CARRYOUTF_DUT == 0))
    $display("All outputs are zero");
else
    $display("Not all outputs are zero");

//Deassert & assert
RSTA_tb = 0;
RSTB_tb = 0;
RSTM_tb = 0;
RSTP_tb = 0;
RSTC_tb = 0;
RSTD_tb = 0;
RSTCARRYIN_tb = 0;
RSTOPMODE_tb = 0;
CEA_tb = 1;
CEB_tb = 1;
CEM_tb = 1;
CEP_tb = 1;
CEC_tb = 1;
CED_tb = 1;
CECARRYIN_tb = 1;
CEOPMODE_tb = 1;

//Path 1
OPMODE_tb = 8'b11011101;
A_tb = 'd20;
B_tb = 'd10;
C_tb = 'd350;
D_tb = 'd25;
BCIN_tb = $random;
PCIN_tb = $random;
CARRYIN_tb = $random;
@(negedge CLK_tb);
@(negedge CLK_tb);
@(negedge CLK_tb);
@(negedge CLK_tb);
```



```
    if ((BCOUT_DUT == 'hf) && (PCOUT_DUT == 'h32) && (P_DUT == 'h32) && (M_DUT ==  
'h12c) && (CARRYOUT_DUT == 0) && (CARRYOUTF_DUT == 0))  
        $display("Path 1 Pass");  
    else  
        $display("Path 1 Error");  
  
//Path 2  
    OPMODE_tb = 8'b00010000;  
    A_tb = 'd20;  
    B_tb = 'd10;  
    C_tb = 'd350;  
    D_tb = 'd25;  
    BCIN_tb = $random;  
    PCIN_tb = $random;  
    CARRYIN_tb = $random;  
    @(negedge CLK_tb);  
    @(negedge CLK_tb);  
    @(negedge CLK_tb);  
    if ((BCOUT_DUT == 'h23) && (PCOUT_DUT == 0) && (P_DUT == 0) && (M_DUT ==  
'h2bc) && (CARRYOUT_DUT == 0) && (CARRYOUTF_DUT == 0))  
        $display("Path 2 Pass");  
    else  
        $display("Path 2 Error");  
  
//Path 3  
    OPMODE_tb = 8'b00001010;  
    A_tb = 'd20;  
    B_tb = 'd10;  
    C_tb = 'd350;  
    D_tb = 'd25;  
    BCIN_tb = $random;  
    PCIN_tb = $random;  
    CARRYIN_tb = $random;  
    @(negedge CLK_tb);  
    @(negedge CLK_tb);  
    @(negedge CLK_tb);  
    if ((BCOUT_DUT == 'ha) && (P_DUT == PCOUT_DUT) && (M_DUT == 'hc8) &&  
(CARRYOUT_DUT == CARRYOUTF_DUT))  
        $display("Path 3 Pass");  
    else  
        $display("Path 3 Error");  
  
//Path 4  
    OPMODE_tb = 8'b10100111;  
    A_tb = 'd5;
```




```
B_tb = 'd6;
C_tb = 'd350;
D_tb = 'd25;
PCIN_tb = 'd3000;
BCIN_tb = $random;
CARRYIN_tb = $random;
@(negedge CLK_tb);
@(negedge CLK_tb);
@(negedge CLK_tb);
if ((BCOUT_DUT == 'h6) && (PCOUT_DUT == 'hfe6ffffec0bb1) && (P_DUT ==
'hfe6ffffec0bb1) && (M_DUT == 'h1e) && (CARRYOUT_DUT == 1) && (CARRYOUTF_DUT ==
1))
    $display("Path 4 Pass");
else
    $display("Path 4 Error");

end
endmodule
```

3) Do File:



run.do - Notepad

File Edit Format View Help

vlib work

```
vlog mux4x1.v
vlog dff_mux.v
vlog dsp_top.v
vlog dsp_tb.v
```

```
vsim -voptargs=+acc work.dsp_tb
```

```
add wave *
```

```
run -all
```

```
#quit -sim
```



4) QuestaSim Snippets:

	Msgs	
CLK_tb	1'h0	
CARRYIN_tb	1'h0	
RSTA_tb	1'h0	
RSTB_tb	1'h0	
RSTM_tb	1'h0	
RSTP_tb	1'h0	
RSTC_tb	1'h0	
RSTD_tb	1'h0	
RSTCARRYIN_tb	1'h0	
RSTOPMODE_tb	1'h0	
CEA_tb	1'h1	
CEB_tb	1'h1	
CEM_tb	1'h1	
CEP_tb	1'h1	
CEC_tb	1'h1	
CED_tb	1'h1	
CECARRYIN_tb	1'h1	
CEOPMODE_tb	1'h1	
A_tb	18'h00005	13524 00014 00005
B_tb	18'h00006	15e81 0000a 00006
BCIN_tb	18'h24ec5	05663 384c5 27277 069f2 24ec5
D_tb	18'h00019	0d609 00019
C_tb	48'h00000000015e	000006... 00000000015e
PCIN_tb	48'h000000000bb8	000046... ffff513d2aa fffe77696de 000000000bb8
OPMODE_tb	8'ha7	65 dd 10 0a a7
BCOUT_DUT	18'h00006	00000 0000a 0000f 00023 0000a 00006
PCOUT_DUT	48'hfe6ffec0bb1	000000000000 000000... 000000... 000000000032 000000000000 fe6ffec... fe6ffec0bb1
P_DUT	48'hfe6ffec0bb1	000000000000 000000... 000000... 000000000032 000000000000 fe6ffec... fe6ffec0bb1
M_DUT	36'h00000001e	0000000000 000000... 00000012c 0000002bc 0000000c8 00000001e
CARRYOUT_DUT	1'h1	
CARRYOUTF_DUT	1'h1	

```
VSIM 35> run
# All outputs are zero
# Path 1 Pass
# Path 2 Pass
# Path 3 Pass
# Path 4 Pass
```



5) Constraint File:

```
create_debug_core u_ila_0 ila
set_property ALL_PROBE_SAME_MU true [get_debug_cores u_ila_0]
set_property ALL_PROBE_SAME_MU_CNT 1 [get_debug_cores u_ila_0]
set_property C_ADV_TRIGGER false [get_debug_cores u_ila_0]
set_property C_DATA_DEPTH 1024 [get_debug_cores u_ila_0]
set_property C_EN_STRG_QUAL false [get_debug_cores u_ila_0]
set_property C_INPUT_PIPE_STAGES 0 [get_debug_cores u_ila_0]
set_property C_TRIGIN_EN false [get_debug_cores u_ila_0]
set_property C_TRIGOUT_EN false [get_debug_cores u_ila_0]
set_property port_width 1 [get_debug_ports u_ila_0/clock]
connect_debug_port u_ila_0/clock [get_nets [list CLK_IBUF_BUFG]]
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe0]
set_property port_width 18 [get_debug_ports u_ila_0/probe0]
connect_debug_port u_ila_0/probe0 [get_nets [list {A_IBUF[0]} {A_IBUF[1]}
{A_IBUF[2]} {A_IBUF[3]} {A_IBUF[4]} {A_IBUF[5]} {A_IBUF[6]} {A_IBUF[7]}
{A_IBUF[8]} {A_IBUF[9]} {A_IBUF[10]} {A_IBUF[11]} {A_IBUF[12]} {A_IBUF[13]}
{A_IBUF[14]} {A_IBUF[15]} {A_IBUF[16]} {A_IBUF[17]}]]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe1]
set_property port_width 36 [get_debug_ports u_ila_0/probe1]
connect_debug_port u_ila_0/probe1 [get_nets [list {M_OBUF[0]} {M_OBUF[1]}
{M_OBUF[2]} {M_OBUF[3]} {M_OBUF[4]} {M_OBUF[5]} {M_OBUF[6]} {M_OBUF[7]}
{M_OBUF[8]} {M_OBUF[9]} {M_OBUF[10]} {M_OBUF[11]} {M_OBUF[12]} {M_OBUF[13]}
{M_OBUF[14]} {M_OBUF[15]} {M_OBUF[16]} {M_OBUF[17]} {M_OBUF[18]} {M_OBUF[19]}
{M_OBUF[20]} {M_OBUF[21]} {M_OBUF[22]} {M_OBUF[23]} {M_OBUF[24]} {M_OBUF[25]}
{M_OBUF[26]} {M_OBUF[27]} {M_OBUF[28]} {M_OBUF[29]} {M_OBUF[30]} {M_OBUF[31]}
{M_OBUF[32]} {M_OBUF[33]} {M_OBUF[34]} {M_OBUF[35]}]]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe2]
set_property port_width 48 [get_debug_ports u_ila_0/probe2]
connect_debug_port u_ila_0/probe2 [get_nets [list {C_IBUF[0]} {C_IBUF[1]}
{C_IBUF[2]} {C_IBUF[3]} {C_IBUF[4]} {C_IBUF[5]} {C_IBUF[6]} {C_IBUF[7]}
{C_IBUF[8]} {C_IBUF[9]} {C_IBUF[10]} {C_IBUF[11]} {C_IBUF[12]} {C_IBUF[13]}
{C_IBUF[14]} {C_IBUF[15]} {C_IBUF[16]} {C_IBUF[17]} {C_IBUF[18]} {C_IBUF[19]}
{C_IBUF[20]} {C_IBUF[21]} {C_IBUF[22]} {C_IBUF[23]} {C_IBUF[24]} {C_IBUF[25]}
{C_IBUF[26]} {C_IBUF[27]} {C_IBUF[28]} {C_IBUF[29]} {C_IBUF[30]} {C_IBUF[31]}
{C_IBUF[32]} {C_IBUF[33]} {C_IBUF[34]} {C_IBUF[35]} {C_IBUF[36]} {C_IBUF[37]}
{C_IBUF[38]} {C_IBUF[39]} {C_IBUF[40]} {C_IBUF[41]} {C_IBUF[42]} {C_IBUF[43]}
{C_IBUF[44]} {C_IBUF[45]} {C_IBUF[46]} {C_IBUF[47]}]]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe3]
set_property port_width 48 [get_debug_ports u_ila_0/probe3]
connect_debug_port u_ila_0/probe3 [get_nets [list {P_OBUF[0]} {P_OBUF[1]}
{P_OBUF[2]} {P_OBUF[3]} {P_OBUF[4]} {P_OBUF[5]} {P_OBUF[6]} {P_OBUF[7]}
{P_OBUF[8]} {P_OBUF[9]} {P_OBUF[10]} {P_OBUF[11]} {P_OBUF[12]} {P_OBUF[13]}
{P_OBUF[14]} {P_OBUF[15]} {P_OBUF[16]} {P_OBUF[17]} {P_OBUF[18]} {P_OBUF[19]}
{P_OBUF[20]} {P_OBUF[21]} {P_OBUF[22]} {P_OBUF[23]} {P_OBUF[24]} {P_OBUF[25]}
{P_OBUF[26]} {P_OBUF[27]} {P_OBUF[28]} {P_OBUF[29]} {P_OBUF[30]} {P_OBUF[31]}
{P_OBUF[32]} {P_OBUF[33]} {P_OBUF[34]} {P_OBUF[35]} {P_OBUF[36]} {P_OBUF[37]}
{P_OBUF[38]} {P_OBUF[39]} {P_OBUF[40]} {P_OBUF[41]} {P_OBUF[42]} {P_OBUF[43]}
{P_OBUF[44]} {P_OBUF[45]} {P_OBUF[46]} {P_OBUF[47]}]]]
```



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```
{PCIN_IBUF[32]} {PCIN_IBUF[33]} {PCIN_IBUF[34]} {PCIN_IBUF[35]} {PCIN_IBUF[36]}  
{PCIN_IBUF[37]} {PCIN_IBUF[38]} {PCIN_IBUF[39]} {PCIN_IBUF[40]} {PCIN_IBUF[41]}  
{PCIN_IBUF[42]} {PCIN_IBUF[43]} {PCIN_IBUF[44]} {PCIN_IBUF[45]} {PCIN_IBUF[46]}  
{PCIN_IBUF[47]}}]  
create_debug_port u_ila_0 probe  
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe9]  
set_property port_width 1 [get_debug_ports u_ila_0/probe9]  
connect_debug_port u_ila_0/probe9 [get_nets [list CARRYOUTF_OBUF]]  
create_debug_port u_ila_0 probe  
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe10]  
set_property port_width 1 [get_debug_ports u_ila_0/probe10]  
connect_debug_port u_ila_0/probe10 [get_nets [list CEA_IBUF]]  
create_debug_port u_ila_0 probe  
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe11]  
set_property port_width 1 [get_debug_ports u_ila_0/probe11]  
connect_debug_port u_ila_0/probe11 [get_nets [list CEB_IBUF]]  
create_debug_port u_ila_0 probe  
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe12]  
set_property port_width 1 [get_debug_ports u_ila_0/probe12]  
connect_debug_port u_ila_0/probe12 [get_nets [list CEC_IBUF]]  
create_debug_port u_ila_0 probe  
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe13]  
set_property port_width 1 [get_debug_ports u_ila_0/probe13]  
connect_debug_port u_ila_0/probe13 [get_nets [list CECARRYIN_IBUF]]  
create_debug_port u_ila_0 probe  
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe14]  
set_property port_width 1 [get_debug_ports u_ila_0/probe14]  
connect_debug_port u_ila_0/probe14 [get_nets [list CED_IBUF]]  
create_debug_port u_ila_0 probe  
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe15]  
set_property port_width 1 [get_debug_ports u_ila_0/probe15]  
connect_debug_port u_ila_0/probe15 [get_nets [list CEM_IBUF]]  
create_debug_port u_ila_0 probe  
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe16]  
set_property port_width 1 [get_debug_ports u_ila_0/probe16]  
connect_debug_port u_ila_0/probe16 [get_nets [list CEOPMODE_IBUF]]  
create_debug_port u_ila_0 probe  
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe17]  
set_property port_width 1 [get_debug_ports u_ila_0/probe17]  
connect_debug_port u_ila_0/probe17 [get_nets [list CEP_IBUF]]  
create_debug_port u_ila_0 probe  
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe18]  
set_property port_width 1 [get_debug_ports u_ila_0/probe18]  
connect_debug_port u_ila_0/probe18 [get_nets [list CLK_IBUF]]  
create_debug_port u_ila_0 probe
```



```
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe19]
set_property port_width 1 [get_debug_ports u_ila_0/probe19]
connect_debug_port u_ila_0/probe19 [get_nets [list RSTA_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe20]
set_property port_width 1 [get_debug_ports u_ila_0/probe20]
connect_debug_port u_ila_0/probe20 [get_nets [list RSTB_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe21]
set_property port_width 1 [get_debug_ports u_ila_0/probe21]
connect_debug_port u_ila_0/probe21 [get_nets [list RSTC_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe22]
set_property port_width 1 [get_debug_ports u_ila_0/probe22]
connect_debug_port u_ila_0/probe22 [get_nets [list RSTCARRYIN_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe23]
set_property port_width 1 [get_debug_ports u_ila_0/probe23]
connect_debug_port u_ila_0/probe23 [get_nets [list RSTD_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe24]
set_property port_width 1 [get_debug_ports u_ila_0/probe24]
connect_debug_port u_ila_0/probe24 [get_nets [list RSTM_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe25]
set_property port_width 1 [get_debug_ports u_ila_0/probe25]
connect_debug_port u_ila_0/probe25 [get_nets [list RSTOPMODE_IBUF]]
create_debug_port u_ila_0 probe
set_property PROBE_TYPE DATA_AND_TRIGGER [get_debug_ports u_ila_0/probe26]
set_property port_width 1 [get_debug_ports u_ila_0/probe26]
connect_debug_port u_ila_0/probe26 [get_nets [list RSTP_IBUF]]
set_property C_CLK_INPUT_FREQ_HZ 300000000 [get_debug_cores dbg_hub]
set_property C_ENABLE_CLK_DIVIDER false [get_debug_cores dbg_hub]
set_property C_USER_SCAN_CHAIN 1 [get_debug_cores dbg_hub]
connect_debug_port dbg_hub/clock [get_nets CLK_IBUF_BUFG]
```



6) Elaboration: Messages:

Tcl Console Messages x Log Reports Design Runs

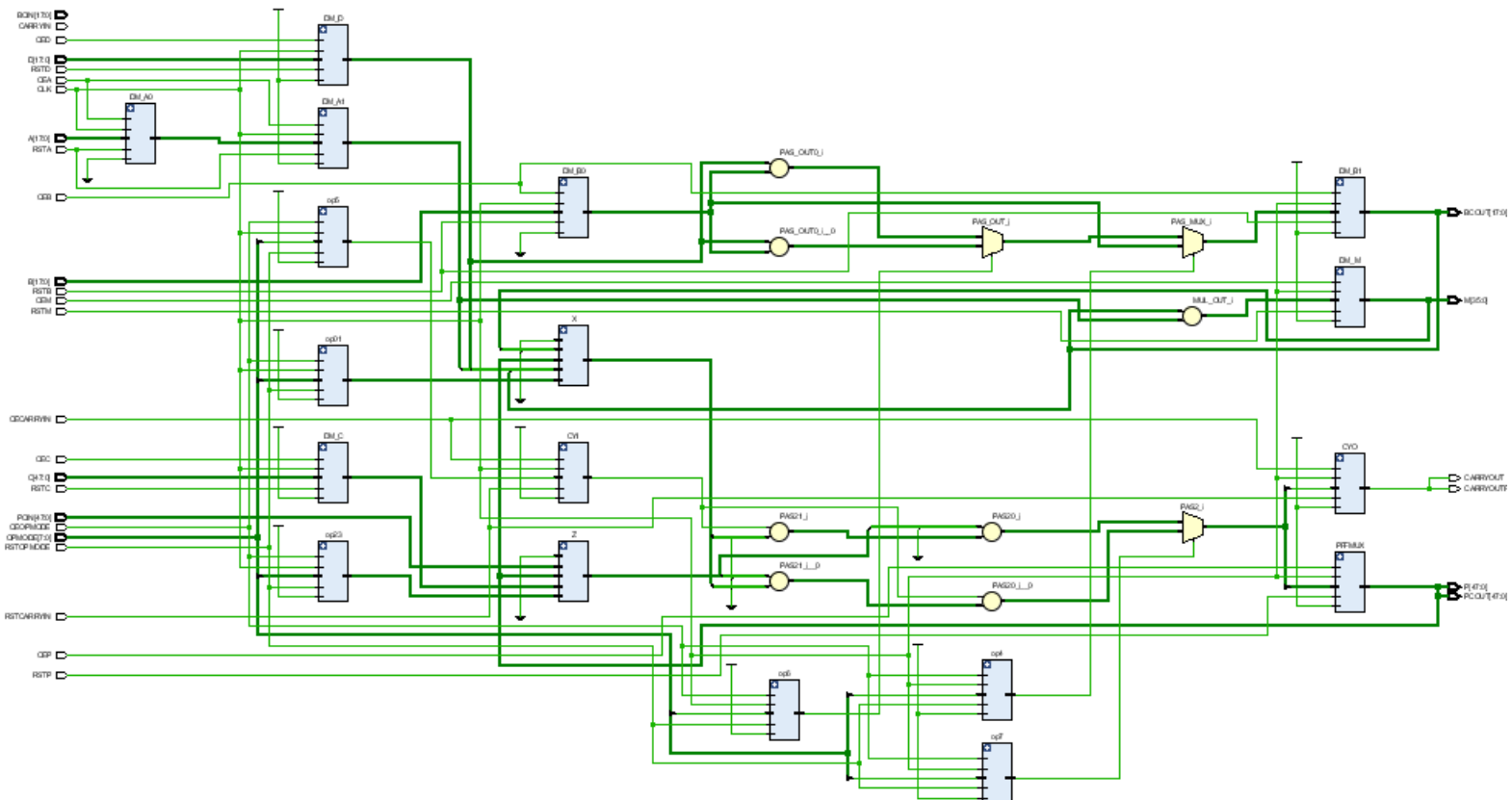
Warning (80) Info (275) Status (496) Hide All

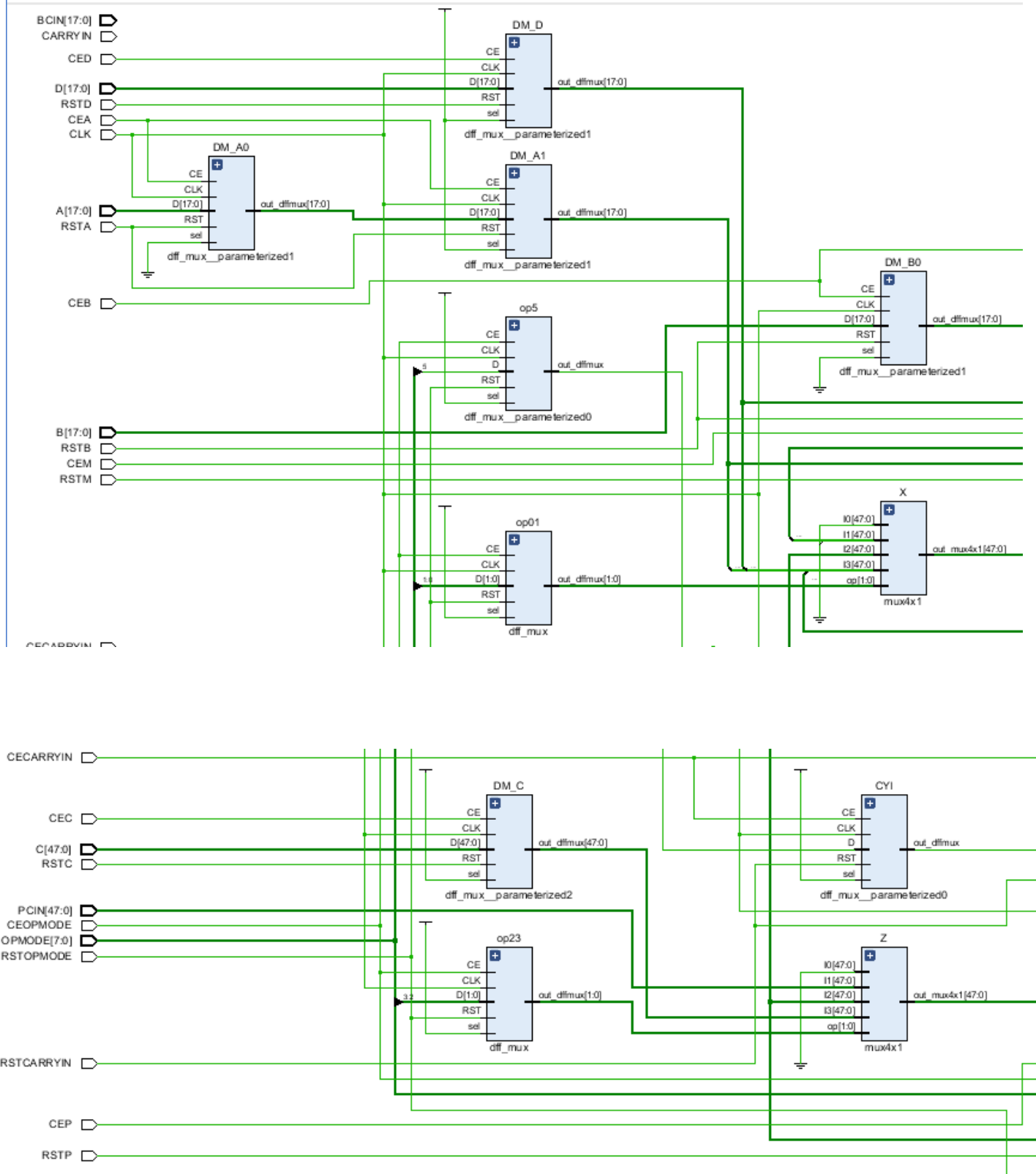
Elaborated Design (19 warnings, 18 infos, 7 status messages)

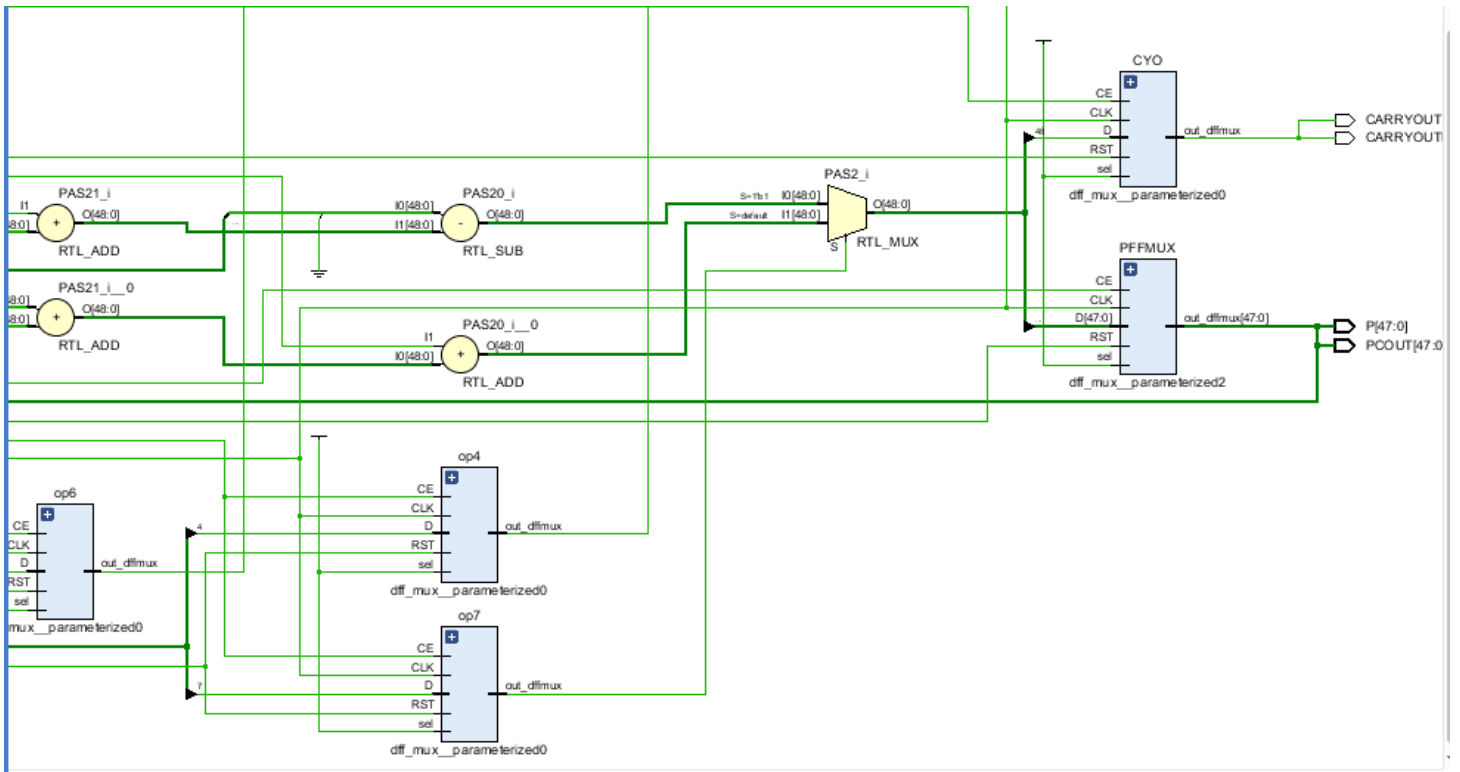
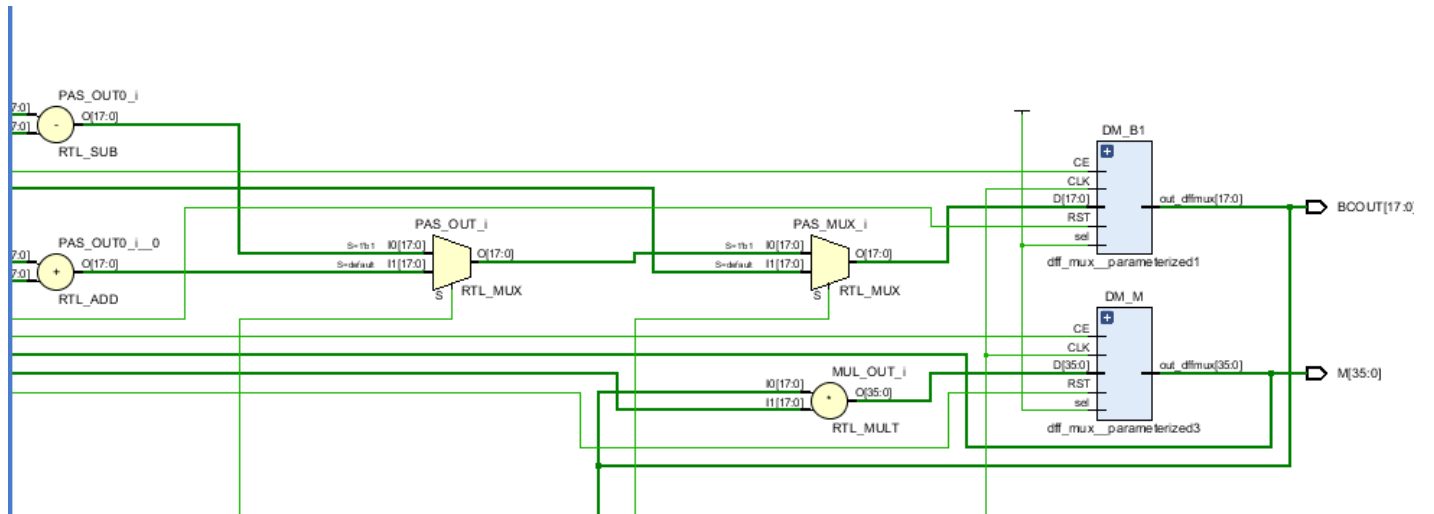
General Messages (19 warnings, 18 infos, 7 status messages)

- [Synth 8-6157] synthesizing module 'dsp_top' [top.v:1] (6 more like this)
- [Synth 8-6155] done synthesizing module 'dff_mux' (1#1) [dff_mux.v:1] (6 more like this)
- [Synth 8-3331] design dsp_top has unconnected port BCIN[17] (18 more like this)
- [Project 1-570] Preparing netlist for logic optimization
- Processing XDC Constraints (6 more like this)
- [Project 1-236] Implementation specific constraints were found while reading constraint file [D:/Mohamed/Koleya/Course/Project1/Constraints_basys3.xdc]. These constraints will be ignored for synthesis but will be used in implementation. Impacted constraints are listed in the file [.Xil/dsp_top_proplmpl.xdc].
Resolution: To avoid this warning, move constraints listed in [.Xil/dsp_top_proplmpl.xdc] to another XDC file and exclude this new file from synthesis with the

Schematic Snippets:









7) Synthesis:

Messages:

SYNTHESIZED DESIGN - xc7a200tffg1156-3 (active)

Tcl Console Messages Log Reports Design Runs Debug

Warning (57) Info (59) Status (19) Hide All

- ✓ Vivado Commands (3 infos, 6 status messages)
 - ✓ General Messages (3 infos, 6 status messages)
 - [IP_Flow 19-234] Refreshing IP repositories
 - [IP_Flow 19-1704] No user IP repositories specified
 - [IP_Flow 19-2313] Loaded Vivado IP repository 'D:/Mohamed/Koleya/Vivado/2018.2/data/ip'.
 - > [Command: synth_design -rtl -name rtl_1 (5 more like this)
- ✓ Synthesis (57 warnings, 50 infos, 11 status messages)
 - > [Command: synth_design -top dsp_top -part xc7a200tffg1156-3 (10 more like this)
 - [Common 17-349] Got license for feature 'Synthesis' and/or device 'xc7a200t'
 - > [Synth 8-6157] synthesizing module 'dsp_top' [top.v:1] (6 more like this)
 - > [Synth 8-6155] done synthesizing module 'dff_mux' (1#1) [dff_mux.v:1] (6 more like this)
 - > [Synth 8-3331] design dsp_top has unconnected port BCIN[17] (37 more like this)
 - [Device 21-403] Loading part xc7a200tffg1156-3
 - [Project 1-236] Implementation specific constraints were found while reading constraint file [D:/Mohamed/Koleya/Course/Project1/Constraints_basys3.xdc]. These constraints will be ignored for synthesis but will be used in implementation. Impacted constraints are listed in the file [.Xil/dsp_top_propimpl.xdc]. Resolution: To avoid this warning, move constraints listed in [Undefined] to another XDC file and exclude this new file from synthesis with the used_in_synthesis property (File Properties dialog in GUI) and re-run elaboration/synthesis.
 - > [Synth 8-5818] HDL ADVISOR - The operator resource <adder> is shared. To prevent sharing consider applying a KEEP on the output of the operator [top.v:48] (1 more like this)
 - > [Synth 8-3886] merging instance 'DM_A0/Q_reg[0]' (FDRE) to 'DM_A1/Q_reg[0]' (17 more like this)
 - > [Synth 8-3332] Sequential element (DM_B0/Q_reg[17]) is unused and will be removed from module dsp_top. (17 more like this)
 - [Synth 8-5842] Cannot pack DSP OPMODE registers because of constant '1' value. Packing the registers will cause simulation mismatch at initial cycle [top.v:60]
 - [Project 1-571] Translating synthesized netlist
 - [Netlist 29-17] Analyzing 207 Unisim elements for replacement
 - [Netlist 29-28] Unisim Transformation completed in 0 CPU seconds
 - > [Project 1-570] Preparing netlist for logic optimization (1 more like this)
 - [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).
 - > [Project 1-111] Unisim Transformation Summary:
No Unisim elements were transformed. (1 more like this)
 - [Common 17-83] Releasing license: Synthesis
 - [Constraints 18-5210] No constraint will be written out.
 - [Common 17-1381] The checkpoint 'D:/Mohamed/Koleya/Course/Project1/Vivado/dsp/dsp.runs/synth_1/dsp_top.dcp' has been generated.
 - [runtcl-4] Executing : report_utilization -file dsp_top_utilization_synth.rpt -pb dsp_top_utilization_synth.pb
 - [Common 17-206] Exiting Vivado at Sat Aug 8 05:26:35 2026...
- ✓ Synthesized Design (6 infos, 2 status messages)
 - ✓ General Messages (6 infos, 2 status messages)



Utilization report:

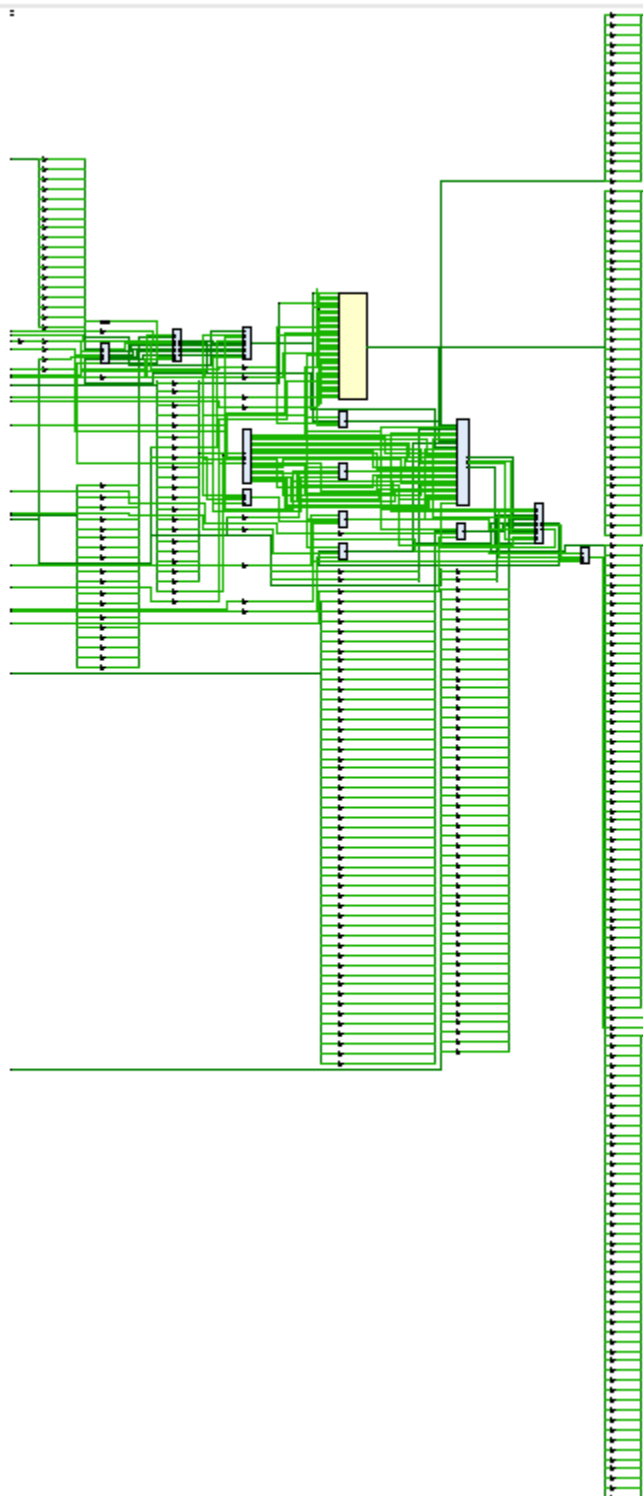
SYNTHESIZED DESIGN - xc7a200tffg1156-3 (active)						
Tcl Console	Messages	Log	Reports	Design Runs	Utilization	Debug
Hierarchy						
Hierarchy						
Summary						
v Slice Logic						
v Slice LUTs (<1%)						
LUT as Logic (<1%)						
v Slice Registers (<1%)						
Register as Flip F						
Memory						
v DSP						
v DSPs (<1%)						
DSP48E1 only						
v IO and GT Specific						
v Bonded IOB (65%)						
IOB Master Pads						
v Clocking						
BUFGCTRL (3%)						
Specific Feature						
Primitives						
Block Devices						
Hierarchy						
Name						
Slice LUTs (134600)						
Slice Registers (269200)						
DSPs (740)						
Bonded IOB (500)						
BUFGCTRL (32)						
v N dsp_top						
CYI (dff_mux__parame...						
CYO (dff_mux__param...						
DM_A1 (dff_mux__par...						
DM_B1 (dff_mux__par...						
DM_C (dff_mux__para...						
DM_D (dff_mux__para...						
op4 (dff_mux__param...						
op5 (dff_mux__param...						
op6 (dff_mux__param...						
op7 (dff_mux__param...						
op01 (dff_mux)						
op23 (dff_mux_4)						
PFFMUX (dff_mux__pa...						

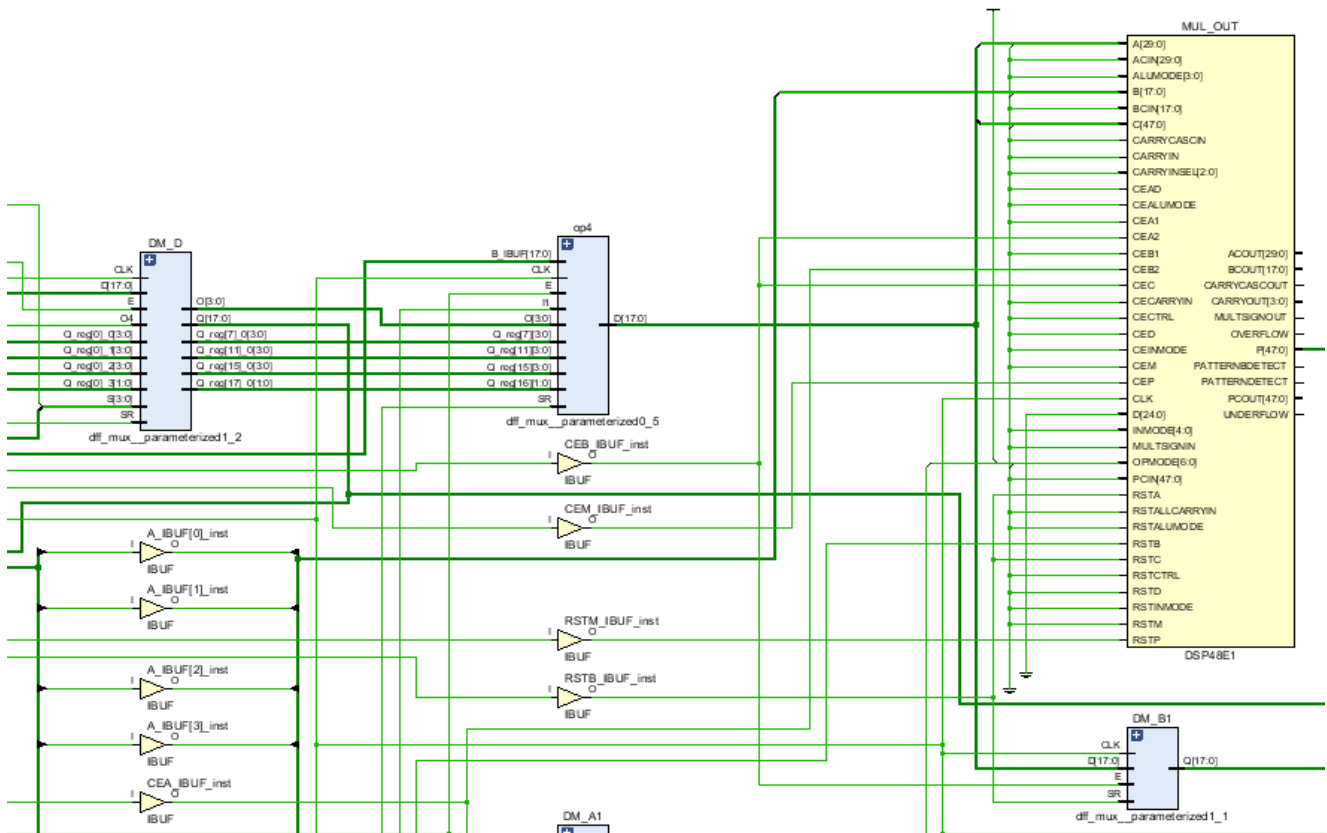
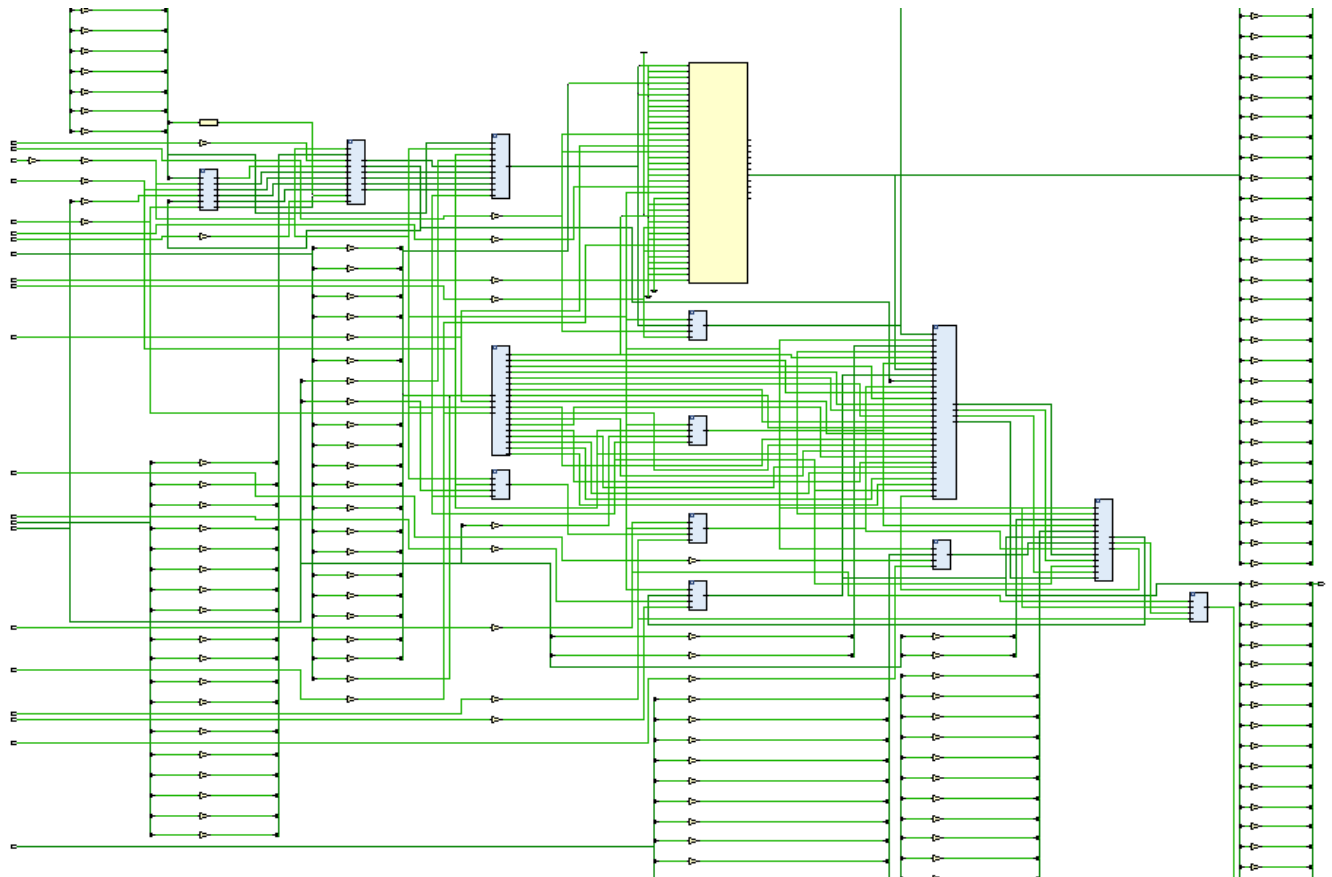
Timing report:

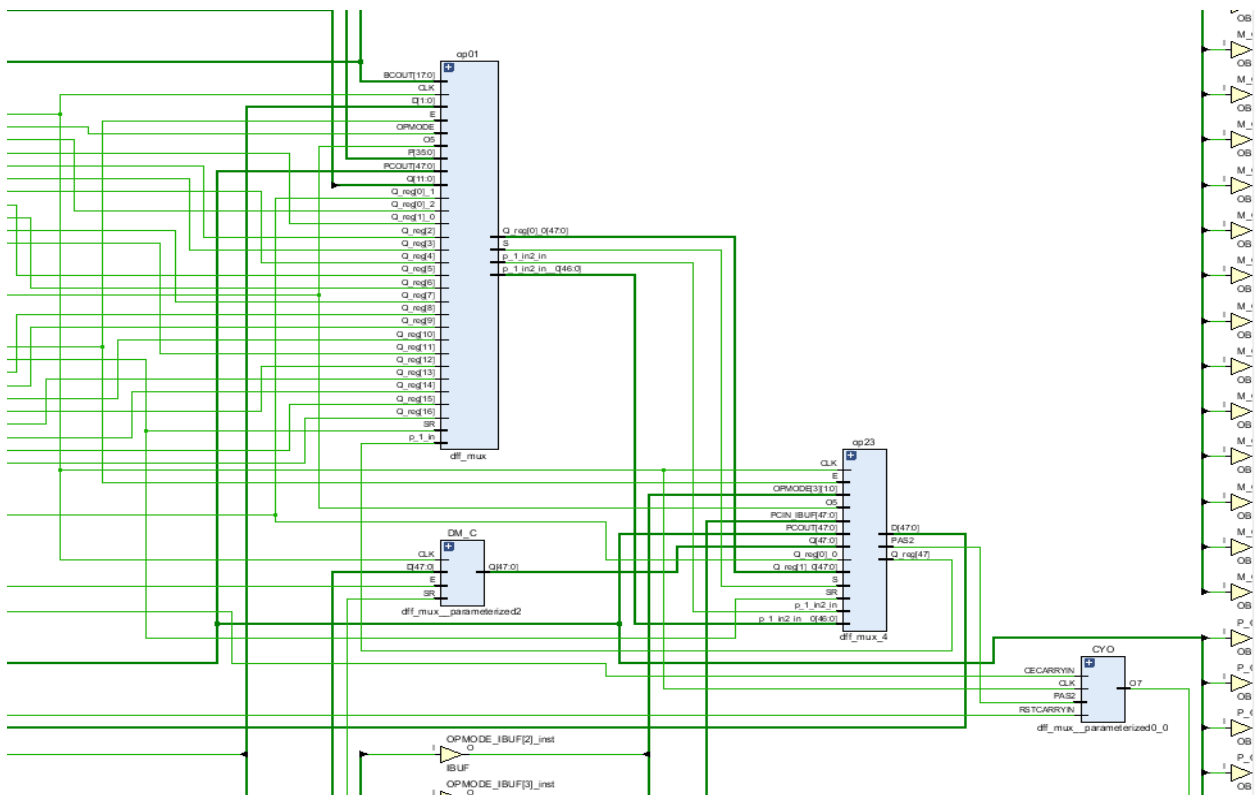
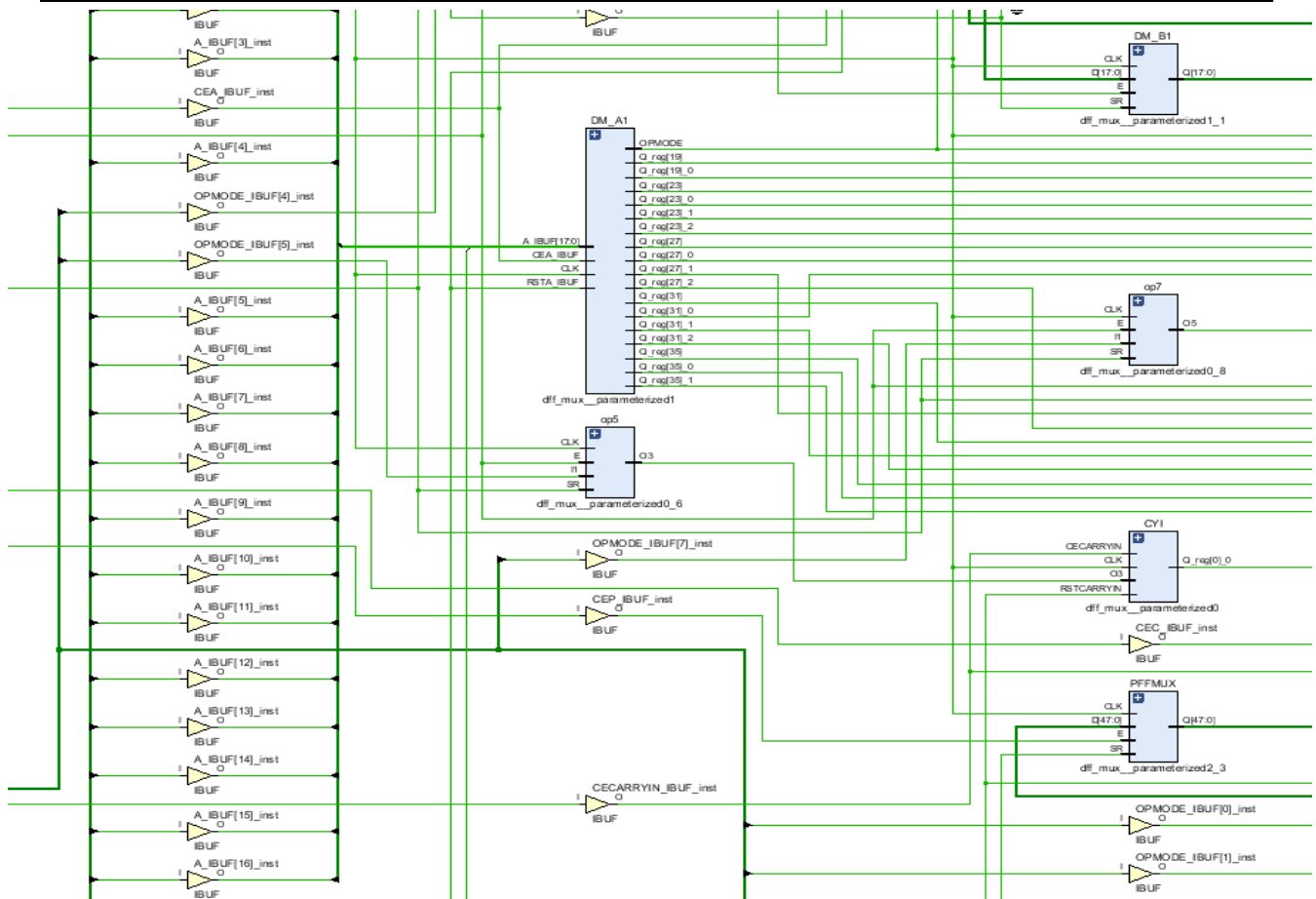
SYNTHESIZED DESIGN - xc7a200tffg1156-3 (active)						
Tcl Console	Messages	Log	Reports	Design Runs	Timing	Utilization
Design Timing Summary						
General Information						
Timer Settings						
Design Timing Summary						
Clock Summary (1)						
> Check Timing (326)						
> Intra-Clock Paths						
Inter-Clock Paths						
Other Path Groups						
User Ignored Paths						
> Unconstrained Paths						
Setup						
Hold						
Pulse Width						
Worst Negative Slack (WNS): 5.168 ns						
Worst Hold Slack (WHS): 0.182 ns						
Worst Pulse Width Slack (WPWS):						
Total Negative Slack (TNS): 0.000 ns						
Total Hold Slack (THS): 0.000 ns						
Total Pulse Width Negative Slack (TPWS):						
Number of Failing Endpoints: 0						
Number of Failing Endpoints: 0						
Number of Failing Endpoints:						
Total Number of Endpoints: 106						
Total Number of Endpoints: 106						
Total Number of Endpoints:						
All user specified timing constraints are met.						



Schematic snippets:









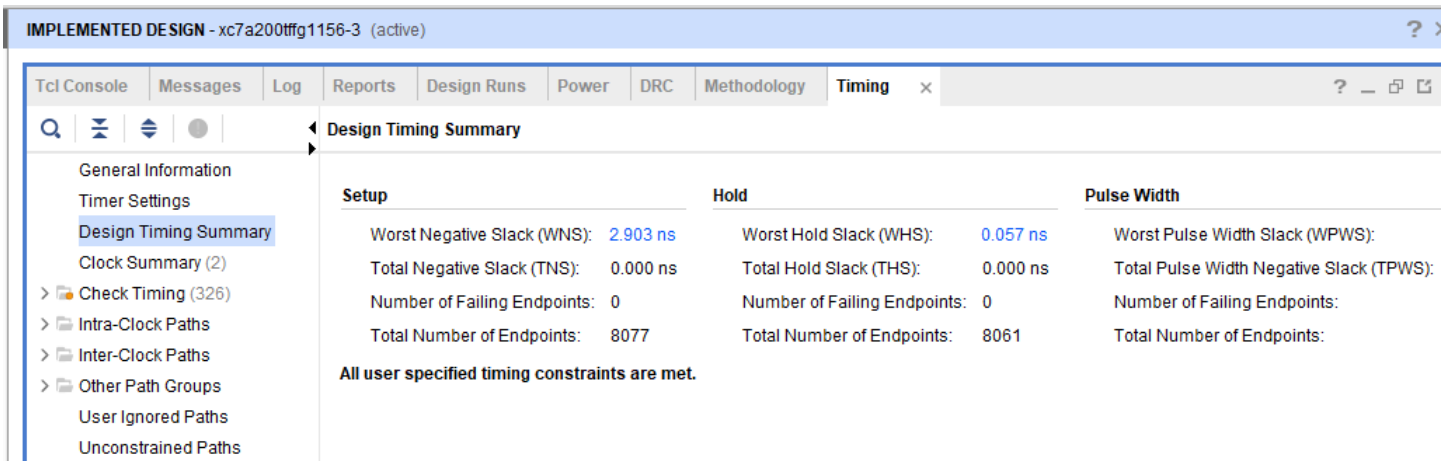
IMPLEMENTED DESIGN - xc7a200tffg1156-3 (active)

Utilization report:

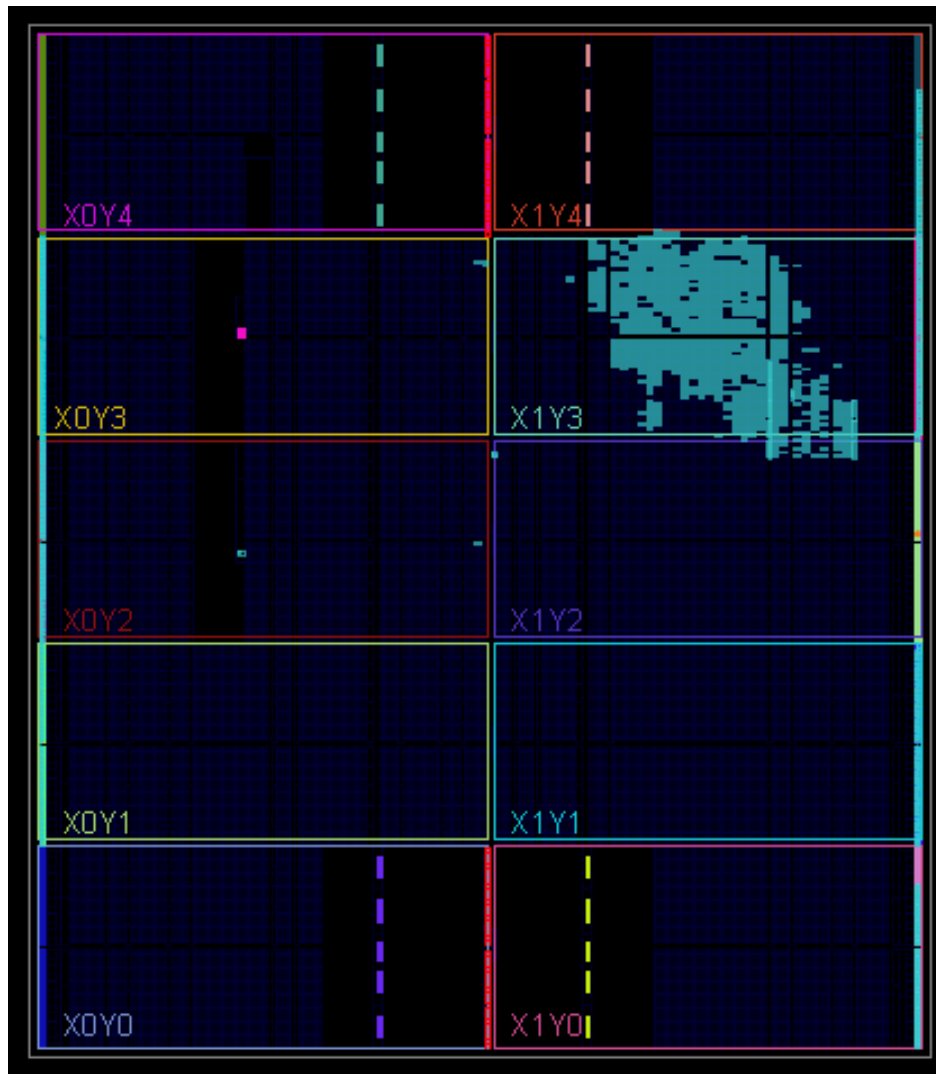
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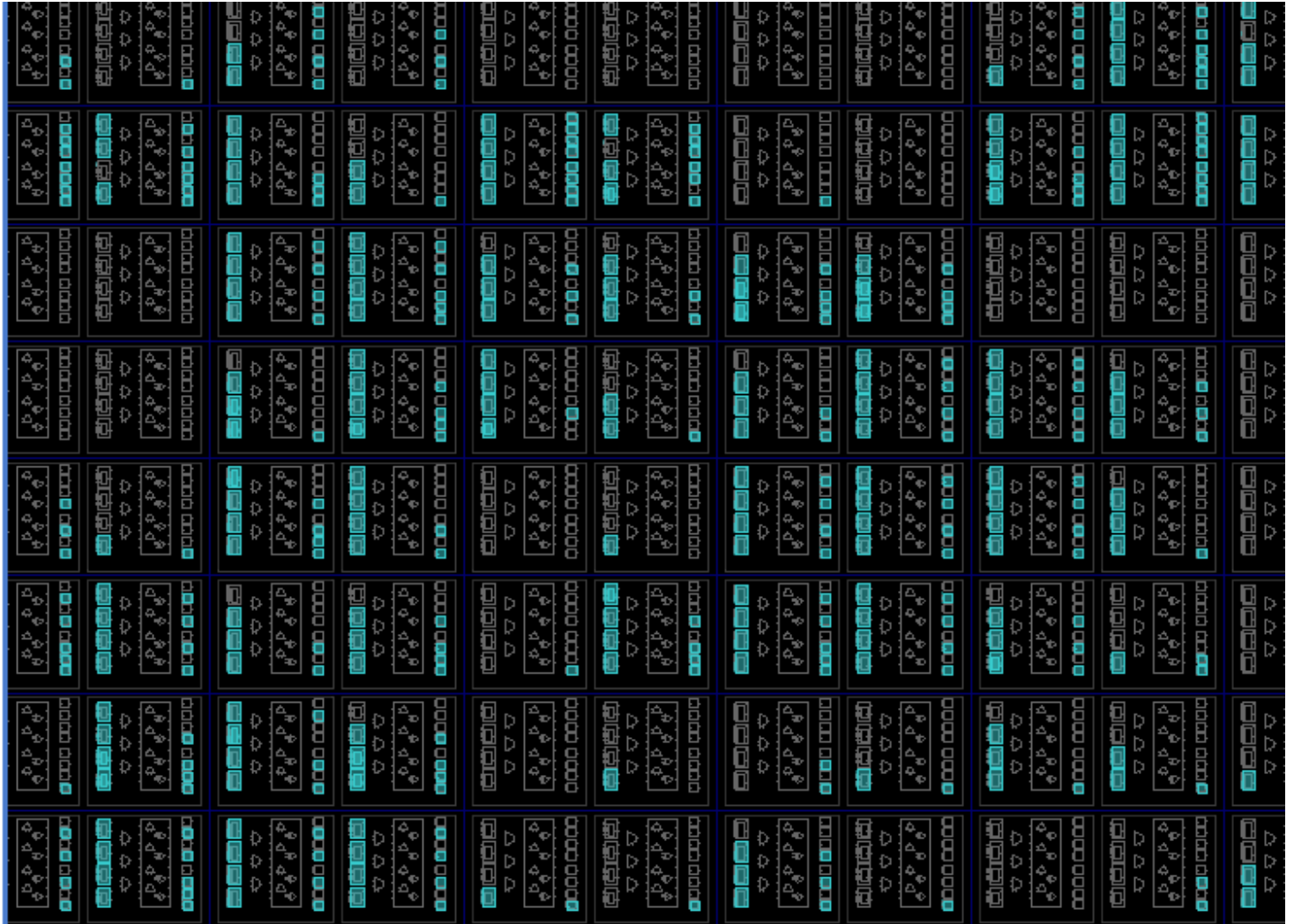


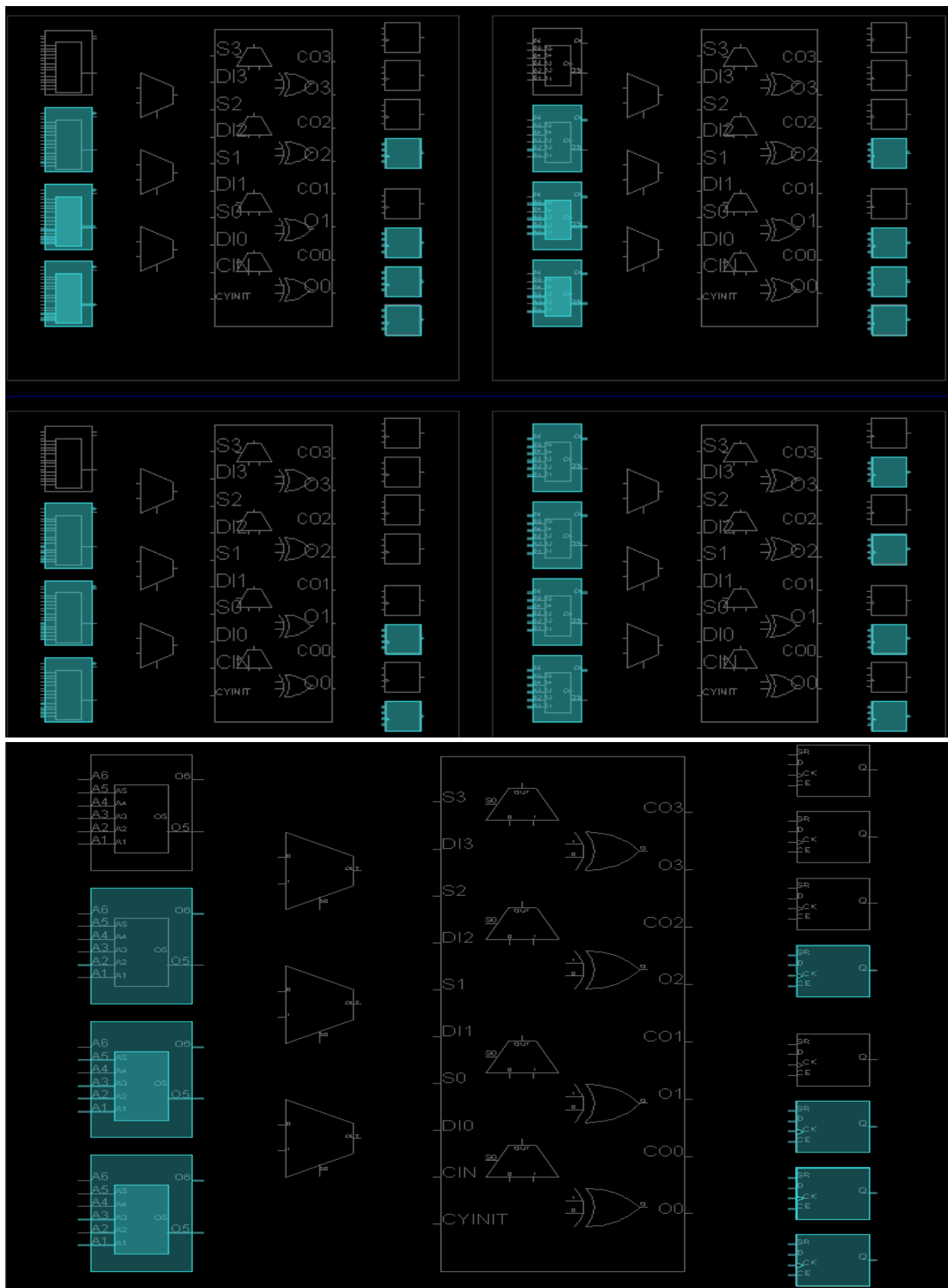
Timing report:



Device Snippets:









9) Linting:

Lint Checks						
Filter: Type here						
Waived Fixed Pending 2 Uninspected Bug Verified Total : 24 Selected : 0						
Severity	Status	Check	Alias	Message	Module	Cate
		mux_select_const		Constant value drives mux select pin. Signal DM_C.out_dffmux, Mod...	dff_mux	Connectiv
		mux_select_const		Constant value drives mux select pin. Signal DM_D.out_dffmux, Mod...	dff_mux	Connectiv
		mux_select_const		Constant value drives mux select pin. Signal DM_M.out_dffmux, Mod...	dff_mux	Connectiv
		mux_select_const		Constant value drives mux select pin. Signal op01.out_dffmux, Modul...	dff_mux	Connectiv
		mux_select_const		Constant value drives mux select pin. Signal op4.out_dffmux, Module ...	dff_mux	Connectiv
		condition_const		Condition expression is a constant. Module dsp_top, File D:/Mohame...	dsp_top	Rtl Desigr
		condition_const		Condition expression is a constant. Module dsp_top, File D:/Mohame...	dsp_top	Rtl Desigr
		parameter_name_duplicate		Same parameter name is used in more than one module. Parameter ...	dff_mux	Nomencla
		line_char_large		Line has more characters than the specified limit. Current Count 120, ...	none	Rtl Desigr
		line_char_large		Line has more characters than the specified limit. Current Count 143, ...	none	Rtl Desigr
		line_char_large		Line has more characters than the specified limit. Current Count 138, ...	none	Rtl Desigr
		line_char_large		Line has more characters than the specified limit. Current Count 111, ...	none	Rtl Desigr
		line_char_large		Line has more characters than the specified limit. Current Count 117, ...	none	Rtl Desigr
		line_char_large		Line has more characters than the specified limit. Current Count 114, ...	none	Rtl Desigr
		line_char_large		Line has more characters than the specified limit. Current Count 122, ...	none	Rtl Desigr
		line_char_large		Line has more characters than the specified limit. Current Count 119, ...	none	Rtl Desigr
		line_char_large		Line has more characters than the specified limit. Current Count 128, ...	none	Rtl Desigr
		line_char_large		Line has more characters than the specified limit. Current Count 116, ...	none	Rtl Desigr
		line_char_large		Line has more characters than the specified limit. Current Count 137, ...	none	Rtl Desigr
		line_char_large		Line has more characters than the specified limit. Current Count 146, ...	none	Rtl Desigr
		line_char_large		Line has more characters than the specified limit. Current Count 126, ...	none	Rtl Desigr
		multi_ports_in_single_line		Multiple ports are declared in one line. Module dff_mux, File D:/Moha...	dff_mux	Rtl Desigr
		multi_ports_in_single_line		Multiple ports are declared in one line. Module mux4x1, File D:/Moha...	mux4x1	Rtl Desigr
		multi_ports_in_single_line		Multiple ports are declared in one line. Module dsp_top, File D:/Moha...	dsp_top	Rtl Desigr